

FUNCTIONAL BLOCK DIAGRAM OF MICROPROCESSOR

A typical microprocessor consists of register array, arithmetic and logic unit (ALU) in association with control unit to process the instruction execution. Almost all the microprocessors are based on the principle of store-program concept. In *store-program concept*, programs or instructions are sequentially stored in the memory locations that are to be executed.

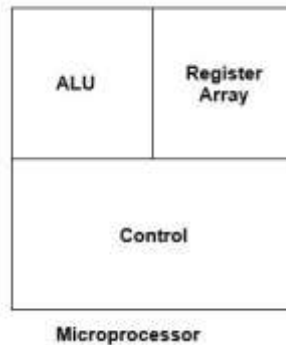


Fig: Functional Block Diagram of Microprocessor

1. ALU – Arithmetic and Logic Unit

The ALU carries out the arithmetic and logic operations on 8-bit words. It can perform arithmetic operations such as addition, subtraction and logical operations such as AND, OR and EX-OR.

The contents of the accumulator and the temporary register are the inputs to the ALU. The ALU result is then stored back in the accumulator. The ALU of a simple less powerful microprocessor such as Intel 8085 does not have multiplication and division instruction.

2. Register Array

Register is a storage unit within the microprocessor used to store the data, address of instruction of any program. Microprocessor contains six general purpose registers it has 8-bit memory and registers are B, C, D, E, H and L register.

To hold 16-bit data a combination of two 8-bit registers can be used. The combination of two 8-bit registers is known as Register Pair (BC, DE and HL). These registers are used to store data temporarily during the execution of the program and are accessible to the user through instructions.

3. Control Unit

The timing and control unit acts as the brain of a computer. It controls all operations of the CPU. It also controls input, output and all other devices connected to the CPU. It generates signals which are required for all the operations to be performed by the CPU and required for the control of input/output and other devices connected to CPU. Such signals are also called timing signals because it has strict relation with clock signal (i.e., time) of the processor.

